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# Dissecting Aneurysm in Cervical Artery Dissection: Insights from the STOP-CAD Study

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Abstract

Background: Cervical artery dissection (CeAD) may result in dissecting aneurysm (DA) formation. We aimed to characterize risk factors and clinical outcomes associated with DA in a large, international CeAD cohort.

Methods: We performed a secondary analysis of the Antithrombotic Treatment for Stroke Prevention in Cervical Artery Dissection (STOP-CAD), an international, multicenter, retrospective cohort. Patients with spontaneous CeAD were assessed for DA at baseline and *de novo* DA formation. Significant covariables in univariable analysis were included in multivariable regression models aiming to identify factors associated with DA formation. Logistic regression and Cox proportional hazards models evaluated the association of DA with ischemic and hemorrhagic outcomes.

Results: Among 4,008 patients with CeAD, 546 (13.6%) had a DA at baseline, and 221/2938 (7.5%) patients developed a *de novo* DA during follow up, of which 172/221 (78.2%) were detected in the first 180 days. Patient with DA had a mean age of 47 years and 383 (49.9%) were women. Baseline or *de novo* DA was associated with history of connective tissue disorder (adjusted odds ratio [aOR] 2.02; 95% CI, 1.22–3.36), fibromuscular dysplasia (aOR 1.69; 95% CI, 1.28–2.25) and multiple vessel dissection (aOR 1.53, 95% CI 1.19-1.98). Lower odds of DA was seen in Hispanic ethnicity (aOR 0.64, 95% CI 0.43-0.97) and ischemic stroke presentation (aOR 0.41, 95% CI 0.34-0.49). Fibromuscular dysplasia was associated with *de novo* DA formation (aOR 2.30; 95% CI, 1.52-3.49). DA was not associated with ischemic stroke (Hazard ratio [HR], 0.71; 95% CI, 0.44–1.16; p=0.173) or intracranial hemorrhage (ICH) (HR 1.09; 95% CI, 0.42–2.84; P=0.86) by day 180.

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Conclusions: DAs are relatively common manifestations of CeAD, typically occurring within six months of CeAD diagnosis. DA was not associated with an increased risk of subsequent ischemic stroke or ICH.

## Nonstandard Abbreviations and Acronyms:

aHR adjusted hazard ratio
aOR adjusted odds ratio
CeAD cervical artery dissection
DA dissecting aneurysm
FMD fibromuscular dysplasia
STOP-CAD Antithrombotic Treatment for Stroke Prevention in Cervical Artery Dissection

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BACKGROUND:

Cervical artery dissection (CeAD) is a leading cause of ischemic stroke, accounting for approximately one-quarter of ischemic strokes in young adults.[1] An intimal tear or rupture of the vasa vasorum results in an intramural hematoma that may extend toward the intima or adventitia. An expansion toward the adventitia can produce a dissecting aneurysm (DA).[1, 2] The prevalence of DA in CeAD ranges from approximately 9% at baseline to 20% within six months, with most patients diagnosed with DA at initial presentation.[1, 3-5] Several risk factors for DA have been reported in small studies, including young age, male sex, migraine history, tobacco use, connective tissue disorders, fibromuscular dysplasia, carotid artery dissection and multiple arterial dissections.[1, 6, 7] However, data on risk factors, *de novo* DA formation, growth, and their associated outcomes remain limited due to small sample size and short follow-up.

The Antithrombotic Treatment for Stroke Prevention in Cervical Artery Dissection (STOP-CAD) registry—a large, multicenter international cohort—provides a unique opportunity to address these knowledge gaps in our understanding of DA in CeAD.[8] Our study aims to analyze the risk factors for DA development and *de novo* DA formation, subsequent DA growth, and clinical outcomes in patients with CeAD and DA.

METHODS:

*Ethics and Data Availability:*

The STOP-CAD study was approved by the institutional review board or ethics committee at each participating site, and all centers complied with local regulatory requirements for retrospective data use. Informed consent was not needed. Aggregate data underlying this analysis

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is available from the corresponding author upon reasonable request and with approval from the STOP-CAD scientific committee.

*Study Design and Population:*

Our study is a secondary analysis of the Antithrombotic Treatment for Stroke Prevention in Cervical Artery Dissection (STOP-CAD) registry, an international, multicenter, retrospective cohort including adults (≥18 years) with confirmed cervical artery dissection (CeAD). Patients were identified from 63 participating centers across North America, South America, Europe, Asia and Africa between 2015 and 2023. Individuals were excluded from the STOP-CAD study if they had major trauma within 4 weeks, chronic, incidental, or iatrogenic dissection or if they presented with subarachnoid hemorrhage due to dissecting pseudoaneurysm. Details about the STOP-CAD registry have been published previously.[8] The study followed the reporting guidelines of Strengthening the Reporting of Observational Studies in Epidemiology.[9]

*Exposure and Definitions:*

A baseline DA was defined as a focal arterial dilatation exceeding the native vessel lumen diameter at the site of dissection, contiguous with the dissected segment, identified on baseline imaging.[1, 4, 10, 11] A *de novo* DA was defined as a new focal enlargement exceeding the native vessel lumen diameter, not present on initial imaging but detected during follow-up.[1, 4, 10] Imaging modalities included computed tomography angiography (CTA), magnetic resonance angiography (MRA), or digital subtraction angiography (DSA), reviewed according to local radiologic standards. Growth of the dissecting aneurysm was defined as any increase in size of the dissecting aneurysm on a follow-up CTA, MRA, or DSA.[4, 10]

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Index ischemic stroke was defined as clinical stroke symptoms or imaging evidence of ischemic stroke corresponding to the dissected artery on initial imaging. Subsequent ischemic stroke was defined as new or worsening neurological symptoms lasting for at least 24 hours or <24 hours but with imaging evidence of new or enlarging acute infarction and referable to the territory of the affected artery. Intracranial hemorrhage (ICH) was defined as a new or worsening intracranial hemorrhage, causing new or worsening neurological symptoms.

Clinical and imaging follow up was at the discretion of the treating physicians, included in the case report form and were reviewed by the STOP-CAD team. Sites were instructed to review all pertinent medical records including available outside hospital records for imaging and outcome abstraction. Sites were queried if there were any discrepancies between the imaging reports and the information recorded and all site principal investigators were queried about every outcome to ensure it is re-reviewed and in keeping with the study definitions.

*Study Cohorts:*

All STOP-CAD patients with available baseline imaging were included in the primary analysis assessing factors associated with DA. Patients without DA at baseline were also evaluated for *de novo* DA formation on follow-up vascular imaging (CTA, MRA, or DSA). Patients with a baseline DA were evaluated for DA growth on follow-up vascular imaging (CTA, MRA, or DSA). For analyses of *de novo DA* formation and DA growth, patients were required to have at least one follow-up vascular imaging study (CTA, MRA, or DSA) after the index CeAD diagnosis.

*Statistical Analysis:*

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Patient demographic, clinical, and imaging variables were compared between groups to identify potential factors associated with DA formation (baseline+de novo). $\chi^2$, Mann-Whitney and Fisher Exact were used as appropriate. Univariable analysis was conducted to preliminarily assess each variable's association with DA formation (Table 1); variables with a *P* value of <0.05 were selected for further analysis using backward stepwise multivariable logistic regression. Model 1 (prespecified clinically relevant) adjusted for age, female sex, race, ethnicity, migraine, hypertension, diabetes, hyperlipidemia, smoking, fibromuscular dysplasia, minor neck trauma, and cervical artery dissection vessel location. Model 2 (data-driven) was derived using stepwise logistic regression with entry and retention thresholds of P<0.05 among candidate predictors. Model 3 (combined) included covariates from Model 1 plus variables retained in Model 2. Variables not retained in the stepwise model are reported as N/A in the corresponding columns of Table 2. Univariable analysis was then repeated for patients with no DA at baseline to identify potential factors associated with *de novo* DA formation. Variables with a *P* value of <0.05 were selected for further analysis using stepwise logistic regression with entry and retention thresholds of P<0.05 among candidate predictors. Variables with p<0.05 were selected for inclusion in the final adjusted binary logistic regression model.

Cox proportional hazards models and Kaplan–Meier survival analyses assessed the association of baseline DA with the risk of subsequent ischemic stroke and intracranial hemorrhagic events up to 180 days. In these time-to-event analyses, patients without baseline DA who developed a *de novo* DA were censored at the time of *de novo* DA detection. Cox proportional hazards models were fit as adjusted models including significant variables differentiating the two groups. Logistic binary regression assessed the association of *de novo* DA and DA growth with the risk of subsequent ischemic stroke and intracerebral hemorrhagic events up to 180 days. For analyses

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of *de novo* DA and outcomes, only events occurring after *de novo* DA detection were counted as outcomes; similarly, for DA growth, only events occurring after the first documentation of growth were counted as outcomes.

Adjusted odds ratios (aORs), hazard ratios (HRs), and 95% confidence intervals (CIs) are reported. Multivariable analyses were performed using complete-case data; patients with missing covariate values were excluded from the corresponding model. All analyses are conducted using Stata SE version 18.0, with statistical significance defined as $p<0.05$.

RESULTS:

*Study Population*

Of 4,023 patients enrolled in the STOP-CAD registry, 4,008 met the inclusion criteria for this analysis (15 patients were excluded due to non-carotid and non-vertebral dissection). Among these, 546 patients (13.6%) had a DA identified on baseline imaging. Of the 3,462 patients without baseline DA, 2938 patients underwent follow up imaging and a total of 221/2938 *de novo* DA (7.5%) were identified in patients with no baseline DA. Furthermore, of the 546 patients with baseline DA, 451 underwent follow-up imaging and 55/451 (12.2%) had DA growth. (Figure 1) The mean age of patients with DA (baseline and *de novo*) was 47 years and 383 (49.9%) were women. (Table 1)

*Factors associated with dissecting aneurysm (baseline and de novo)*

In univariable analyses, pertinent factors associated with an increased odds of DA were female sex (49.9% vs. 43.3%; p=0.001), history of migraine (22.2% vs. 15.3%; p<0.001), fibromuscular dysplasia (15.4% vs. 6.5%; p<0.001), history of connective tissue disorder (4.4% vs. 1.5%; p<0.001), presentation with headache (54.8% vs. 47.1%; p<0.001), neck pain (34.3% vs. 30.3%;



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p=0.033), and imaging findings of multiple extracranial arterial dissections (19% vs. 9.2%; p<0.001; Table 1). Pertinent factors associated with a lower rate of DA were Hispanic ethnicity (5.3% vs. 10.1%; P<0.001), active smoking (15.4% vs. 20.4%; p=0.002), single extracranial vertebral dissection (25.4% vs. 38.2%; p<0.001) and ischemic stroke presentation (45.9% vs. 69.5%, p<0.001) (Table 1). The remaining variables are listed in Table 1.

In multivariable analysis, connective tissue disorder (adjusted odds ratio [aOR], 2.02; 95% CI, 1.22–3.36; p=0.007), fibromuscular dysplasia (aOR, 1.69; 95% CI, 1.28–2.25; p<0.001) and multiple vessel dissection (aOR 1.53, 95% CI 1.19-1.98; p=0.001) were independently associated with DA (Table 2). Furthermore, lower odds of DA was seen in Hispanic ethnicity (aOR 0.64, 95% CI 0.43-0.97; p=0.034) and ischemic stroke presentation (aOR 0.41, 95% CI 0.34-0.49; p<0.001).

*De Novo Dissecting Aneurysm*

Among 3,462 patients without baseline DA, 2938 (84.8%) patients underwent follow up imaging. *De novo* DA developed in 221/2938 (7.5%) of those without baseline DA who underwent follow up imaging (Figure 2a). The timing of *denovo* DA diagnosis was not recorded in 1 patient. Most cases were identified within the first 180 days: 127/220 (57.5%) by 90 days and 172/220 (78.2%) by 180 days (Figure 2b). The median (IQR) time to diagnosis of a de novo DA was 73 (21-159) days. Subsequent follow-up imaging was performed in 153 (70%) patients with *de novo* DA (Median [IQR]: 2 follow-up images [1-3]).

In univariable analyses restricted to only patients within the follow-up cohort and without baseline DA, factors associated with an increased risk of *de novo* DA formation were fibromuscular dysplasia (18.1% vs. 8.2%; p<0.001), history of connective tissue disorder (4.5%

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vs. 2.1%; p=0.02), single extracranial carotid dissection (59.7% vs. 49.6%; p=0.005) (Table 3). Factor associated with a lower rate of *de novo* DA was active smoker status (12.7% vs. 18.7%; p=0.025), Hispanic ethnicity (3.8% vs. 9.2%, p = 0.01), single extracranial vertebral dissection (23.1% vs. 34.4%; p=0.001), and presentation with neck pain (25.8% vs. 34.3%, p = 0.01) (Table 3). The rest of the variables are listed in Table 3.

In multivariable regression analysis restricted to only patients without baseline DA and with at least one follow-up imaging study and including with entry and retention thresholds of P<0.05 among candidate predictors, fibromuscular dysplasia (FMD) was associated with increased odds of *de novo* DA formation (aOR, 2.30; 95% CI, 1.52–3.49; p<0.001; Table 4). Furthermore, neck pain at presentation (aOR, 0.57; 95% CI, 0.40-0.82; p=0.002, Hispanic ethnicity (aOR 0.41, 95% CI 0.19-0.88, p=0.022) and active smoking status (aOR 0.62, 95% CI 0.39-0.97, p=0.038) were associated with lower odds of *de novo* DA formation.

*Dissecting Aneurysm Growth*

Patients with DA at baseline (n=546) were followed for DA growth. Of these, 451 (82.6%) underwent follow-up imaging. DA growth was noted in 55/451 (12.2%; mean age: 46 years; 44% female) of the 546 patients with a baseline DA. Of the 55 patients with DA growth, 49 (89.1%) had it diagnosed within the 180 days. No measured variable was independently associated with DA growth. (Table S1)

*Clinical Outcomes*

There was no significant difference in the risk of subsequent ischemic stroke ([aHR] 0.71, 95% CI, 0.44-1.16; p=0.17) or intracranial hemorrhage ([aHR] 1.09, 95% CI, 0.42-2.84; p=0.86) between patients with and without DA at baseline (Figure 3a and 3b). *De novo* DA was also not

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associated with an increased odds of subsequent ischemic stroke ([OR] 0.50, 95% CI, 0.23-1.09; p=0.08) or intracranial hemorrhage ([OR] 0.40, 95% CI, 0.05–2.91; p=0.36). DA growth was not associated with a significant increase in the odds of subsequent ischemic stroke ([OR] 1.57; 95% CI, 0.44–5.66; p=0.49) or intracranial hemorrhage ([OR] 2.43, 95% CI 0.25–23.74, p=0.45). DA at baseline and *De novo* DA were not associated with an increased rate of mortality (No DA 2% vs. DA 1.2%; p=0.13) during follow up.

DISCUSSION:

In this post hoc analysis of the STOP-CAD study, DA were present in approximately 13.6% of patients with CeAD at baseline. *De novo* DA was noted in 7.5% of patients with no DA at baseline. Only 12.2% of those with baseline DAs had growth of the DA on follow up imaging. Baseline DA, *de novo* DA, and DA growth were not associated with evidence of increased risk of subsequent ischemic stroke or intracranial hemorrhage within 180 days. These findings suggest that the presence of DA, its subsequent growth or *de novo* formation of DA may not necessitate intensified antithrombotic therapy or imaging surveillance from a stroke prevention standpoint.

In prior studies, DAs were found in 9.1–20% of CeAD cases at initial presentation.[1, 3-5, 12] Similar to prior studies, symptoms related to DA included headache, neck pain, Horner's syndrome and amaurosis fugax.[2, 13-16] However, it is important to note that these symptoms could be related to CeAD and not necessarily a consequence of DA. Our study did not find a higher prevalence of ischemic stroke at initial presentation in patients with DA, similar to prior studies.[3, 5] Contrary to our study, previous studies did not show any racial or ethnic differences in DA formation in CeAD patients.[1-3, 5, 17] We found an association between non-Hispanic ethnicity and DA

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formation in CeAD. Previously published studies included patients from Europe and North America with limited racial and ethnic diversity in patient cohorts as compared to STOP-CAD which enrolled a globally diverse cohort highlighting the impact of large diverse cohorts on detecting racial and ethnic differences. Recent reports have suggested an increase in overall incidence of CeAD specially in black and hispanic populations.[18, 19] However, race and ethnicity specific data pertaining to DA in CeAD is limited.

Similar to our study, multiple vessel dissections, connective tissue disorders and fibromuscular dysplasia are also associated with an increased risk of DA.[1, 4, 6, 7, 20] Recent reports have suggested connective tissue aberrations and poor wound healing in patients with CeAD. It is possible that this underlying arteriopathy when exposed to hemodynamic stress leads to CeAD complicated by DA due to poor containment of intramural hematoma.[21]

The rate of *de novo* DA formation was 7.5% (221) and most frequently occurred within the first 180 days (172/220, 78.2%), which is consistent with prior studies.[1, 3, 5, 7] Multiple dissections, arterial tortuosity, and underlying connective tissue or vascular disorders have been identified as risk factors for *de novo* DA in CeAD.[1, 7] However, our study found that fibromuscular dysplasia was the only predictor of *de novo* DA formation. Our study showed a lower rate of *de novo* DA formation in active smokers. Smoking is not an independent risk factor for CeAD and the association with DA is unclear.[1] However, it is important to note that smoking is associated with higher risk of ischemic stroke in patients with CeAD.[1] Further investigation is needed to evaluate the association of smoking with DA formation in CeAD.

Most DA remained stable or decreased in size. We detected DA growth in 55 (12.2%) patients with DA at baseline, of which 48 (8.8%) patients showed growth within 180 days of onset, consistent with earlier reports.[4, 10, 22-24] DA in the vertebral artery, presence of intramural



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hematoma on imaging, large initial dissecting aneurysm size, smoking, trauma, and arterial tortuosity have been reported as predictors of DA growth.[1, 3, 7, 13] We did not find any independent predictors of DA growth in our analysis.

Multiple previous studies demonstrated no impact of DA on the risk of recurrent ischemic stroke in patients with CeAD, consistent with our findings.[1, 3, 4, 10] *De novo* DA formation and DA growth showed no evidence of increased risk of ischemic stroke in our study, similar to previous studies.[3, 4, 10]

Growth in DA at presentation and *de novo* DA formation raises the question of screening and imaging surveillance. In the absence of new symptoms, routine screening and long-term imaging surveillance for DAs in CeAD is not supported by current evidence.[1, 12, 25, 26] Our study findings of a relatively benign course of DA despite growth re-affirms this. Moreover, most patients with CeAD and DA patients can be treated with antithrombotics similar to those with CeAD without a DA. It is important to note that imaging follow-up was at the discretion of the treating physician at each STOP-CAD site and might have been influenced by the initial symptoms and risk factors.

As endovascular therapies continue to advance and broaden in scope, elective interventions for DA have become increasingly common, possibly among patients who might otherwise achieve favorable outcomes with medical management alone.[22] Our study results of no impact of DA as it pertains to subsequent ischemic stroke and intracerebral hemorrhage, may suggest that endovascular or surgical intervention is best reserved on an individual case basis where the DA causes significant mass effect or compressive symptoms, or is the culprit for recurrent ischemic events despite maximal medical therapy, rather than being performed for routine management based solely on the presence of DA.[1] Prior studies indicate that endovascular intervention in appropriately chosen patients can generally be performed safely.[27, 28]



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LIMITATIONS:

Our study has several limitations. Multicenter retrospective registries are subject to inherent selection and information biases. Diagnosis relied on clinical documentation and imaging interpretation across participating centers without central adjudication of images, introducing potential variability in DA detection influenced by concomitant diagnoses. Imaging data did not include information about arterial tortuosity and initial DA size. Follow-up imaging intervals were not standardized, which may have influenced observed rates of de novo DA formation as well as DA growth. Moreover, the wide confidence intervals indicate limited power for *de novo* DA and DA growth analyses. Given retrospective design, our study also lacks in-depth multidimensional interaction analysis between DA formation/growth and treatment strategy. Despite these limitations, the large, international cohort and systematic data collection strengthen the generalizability of our findings.

CONCLUSION:

In a large, multinational cohort of patients with CeAD, DA was not associated with an elevated risk of subsequent ischemic stroke or intracranial hemorrhage. Our results question the need for routine surveillance imaging or intensified therapy based solely on the presence, growth or *de novo* formation of DA. Future prospective studies integrating well defined case definitions, collection of genetic profile, standardized clinical, advanced imaging and medical management follow up at regular pre-defined intervals may help identify the mechanisms underlying DA formation and identify patient subgroups who may benefit from tailored monitoring or intervention.

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Conflicts of Interest:

Dr Leker reports compensation from Janssen Biotech for other services; compensation from Filterlex for consultant services; compensation from Boehringer Ingelheim for other services; grants from Horizon 2020 Framework Programme; compensation from Biogen for other services; compensation from Abbott Diabetes Care for other services; compensation from Bayer for other services; compensation from Ischemaview for other services; and compensation from Medtronic for other services. Dr Engelter reports travel support from Boehringer Ingelheim; grants from Daiichi-Sankyo; travel support from Pfizer; grants from Daiichi-Sankyo; and travel support from Bayer. Dr Heldner reports grants from Swiss National Science Foundation; grants from Swiss Heart Foundation; and grants from SITEM Research Funds. Dr Antonenko reports grants from Medtronic; grants from Swiss National Science Foundation; and grants from Gottfried und Julia Bangerter-Rhyner-Stiftung. Dr Siegler reports compensation from Bayer for consultant services; grants from Medtronic; compensation from AstraZeneca for other services; and grants from Philips. Dr Rothstein reports grants from American Heart Association.
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services. Dr Simpkins reports compensation from Up to Date for other services; compensation from NINDS for data and safety monitoring services; and grants from Bristol-Myers Squibb Foundation. Dr Stretz reports grants from American Heart Association to other; grants from Massachusetts General Hospital to other; and grants from Duke University Medical Center to other. Dr Yaghi reports compensation from Medtronic USA, Inc. for other services. Dr Nguyen reports compensation from American Stroke Association for other services; compensation from Medtronic for consultant services; compensation from brainomix for consultant services; compensation from Route 92 Medical, Inc. for consultant services; compensation from Genentech for other services; and compensation from Bayer for consultant services. All the other authors report no conflict of interest

**Supplemental Materials:**

STROBE Checklist

Supplementary Table 1 (Table S1): Univariable Analysis of Preselected Variables Across Patients With or Without Dissecting Aneurysm growth (excluding patients without follow up imaging after initial diagnosis of dissecting aneurysm)

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Table 1: Univariate Analysis of Preselected Variables Across Patients With or Without Dissecting Aneurysm (Baseline and *Denovo*)

| | All Patients (N=4008) | Patients with no dissecting aneurysm (N=3241) | Patients with dissecting aneurysm (N=767) | P-value |
|---|---|---|---|---|
| Demographic | | | | |
| Age, years, median [IQR] | 46[38-56] | 47 [38-56] | 46 [37-55] | 0.45 |
| Female, n/N(%) | 1786/4008(44.6%) | 1403/3241(43.3%) | 383/767(49.9%) | 0.001 |
| *Race* | | | | 0.049 |
| White | 2945/4008(73.5%) | 2353/3241(72.6%) | 592/767(77.2%) | |
| Black | 240/4008(6.0%) | 202/3241(6.2%) | 38/767(5.0%) | |
| Asian | 147/4008(3.7%) | 118/3241(3.6%) | 29/767(3.8%) | |
| Other or unknown race | 676/4008(16.9%) | 568/3241(17.5%) | 108/767(14.1%) | |
| Hispanic ethnicity | 335/3631(9.2%) | 298/2938(10.1%) | 37/693(5.3%) | <0.001 |
| Comorbidities | | | | |
| Migraine | 667/4008(16.6%) | 497/3241(15.3%) | 170/767(22.2%) | <0.001 |
| Hypertension | 1416/4008(35.3%) | 1148/3241(35.4%) | 268/767(34.9%) | 0.80 |
| Diabetes | 337/4008(8.4%) | 285/3241(8.8%) | 52/767(6.8%) | 0.07 |
| Active smoking | 778/4008(19.4%) | 660/3241(20.4%) | 118/767(15.4%) | 0.002 |
| Fibromuscular Dysplasia | 328/4003(8.2%) | 210/3237(6.5%) | 118/766(15.4%) | <0.001 |
| Connective tissue disorder | 83/4008(2.1%) | 49/3241(1.5%) | 34/767(4.4%) | <0.001 |
| Presenting Symptoms | | | | |
| Headache | 1948/4008(48.6%) | 1528/3241(47.1%) | 420/767(54.8%) | <0.001 |
| Neck Pain | 1246/4008(31.1%) | 983/3241(30.3%) | 263/767(34.3%) | 0.03 |
| Minor neck trauma | 892/4007(22.3%) | 722/3240(22.3%) | 170/767(22.2%) | 0.94 |
| Presenting AIS | 2602/4006(65.0%) | 2250/3239(69.5%) | 352/767(45.9%) | <0.001 |
| Imaging | | | | |

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| Dissection location | | | | |
|---|---|---|---|---|
| Single extracranial carotid dissection | 1983/4008(49.5%) | 1582/3241(48.8%) | 401/767(52.3%) | 0.08 |
| Single extracranial vertebral dissection | 1432/4008(35.7%) | 1237/3241(38.2%) | 195/767(25.4%) | <0.001 |
| Single intracranial carotid or vertebral dissection | 150/4008 (3.7%) | 125/3241 (3.9%) | 25/767 (3.3%) | 0.43 |
| Multiple arterial dissection | 443/4008(11.1%) | 297/3241(9.2%) | 146/767(19.0%) | <0.001 |

N=Number of Patients; IQR= interquartile range; AIS=Acute Ischemic Stroke

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Table 2: Unadjusted and Adjusted Multivariable logistic regression comparing Odds Ratios of dissecting aneurysm (Baseline and *Denovo*) for Statistically Significant Variables Identified

| **Variable** | **Unadjusted OR (95% CI, p-value)** | **Adjusted OR (95% CI, p-value)** | | |
|---|---|---|---|---|
| | | Model 1 (prespecified clinically relevant) | Model 2 (Stepwise) | Model 3 (Combined variables from model 1 and model 2) |
| Age | 0.99 (0.99-1.004, 0.45) | 1.003 (0.99-1.01, 0.42) | N/A | 0.99 (0.99-1.007, 0.89) |
| Female | 1.30 (1.11-1.53, 0.001) | 1.07 (0.89-1.28, 0.44) | N/A | 0.95 (0.78-1.14, 0.60) |
| Race | | | | |
| White race | 1.27 (1.06-1.53) | N/A (used as baseline) | N/A | N/A (used as baseline) |
| Black race | 0.78 (0.54-1.11) | 0.82 (0.57-1.19, 0.31) | N/A | 0.91 (0.62-1.33, 0.64) |
| Asian race | 1.04 (0.68-1.57) | 1.020 (0.66-1.56, 0.92) | N/A | 1.13 (0.73-1.75, 0.57) |
| Other or unknown race | 0.771 (0.61-0.96) | 0.824 (0.56-1.20, 0.31) | N/A | 0.825 (0.56-1.20, 0.32) |
| Hispanic ethnicity | 0.50 (0.35-0.71, <0.001) | 0.570 (0.38-0.85, 0.007) | 0.58 (0.40-0.840, 0.004) | 0.64 (0.42-0.96, 0.03) |
| Migraine | 1.57 (1.29-1.91, <0.001) | 1.44 (1.15-1.79, 0.001) | 1.23 (0.99-1.53, 0.05) | 1.24 (0.99-1.55, 0.05) |
| Hypertension | 0.97 (0.83-1.15, 0.80) | 0.97 (0.79-1.20, 0.83) | N/A | 0.99 (0.80-1.22, 0.94) |
| Diabetes | 0.75 (0.55-1.02, 0.07) | 0.89 (0.63-1.24, 0.49) | N/A | 1.00 (0.71-1.40, 0.99) |
| Hyperlipidemia | 0.86 (0.71-1.04, 0.13) | 0.94 (0.75-1.18, 0.63) | N/A | 0.96 (0.76-1.21, 0.73) |
| Active smoking | 0.71 (0.57-0.88, 0.002) | 0.74 (0.58-0.93, 0.01) | 0.80 (0.63-1.014, 0.06) | 0.79 (0.63-1.01, 0.06) |
| Connective tissue disorder | 3.02 (1.93-4.71, <0.001) | N/A | 2.02 (1.22-3.35, 0.006) | 2.02 (1.21-3.36, 0.007) |
| Fibromuscular dysplasia | 2.62 (2.06-3.34, <0.001) | 1.95 (1.48-0-2.56, <0.001) | 1.67 (1.26-2.21, <0.001) | 1.69 (1.27-2.24, <0.001) |
| Minor neck trauma | 0.99 (0.82–1.20, 0.94) | 0.91 (0.74-1.13, 0.42) | N/A | 0.92 (0.74-1.13, 0.45) |
| Vessel Location | | | | |



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| | | | | |
|---|---|---|---|---|
| Single extracranial carotid dissection | 1.14 (0.98–1.34, 0.08) | N/A | N/A | N/A |
| Single extracranial vertebral dissection | 0.55 (0.46–0.65, <0.001) | N/A | 0.66 (0.54-0.81, <0.001) | 0.66 (0.54-0.81, <0.001) |
| Multiple vessel dissection | 2.33 (1.87–2.89, <0.001) | 1.87 (1.47-2.38, <0.001) | 1.52 (1.19-1.95, 0.001) | 1.53 (1.18-1.97, 0.001) |
| Presenting AIS | 0.37 (0.31–0.43, <0.001) | N/A | 0.41 (0.34-0.49, <0.001) | 0.40 (0.34-0.48, <0.001) |

OR= Odds Ratio; CI=Confidence Interval; AIS=Acute Ischemic Stroke

Model 1: adjusted for prespecified clinically relevant variables of age, female sex, race, Hispanic ethnicity, history of migraines, history of hypertension, history of diabetes, hyperlipidemia, active smoking, fibromuscular dysplasia, minor neck trauma, multiple cervical artery vessel dissection.

Model 2: adjusted for Hispanic ethnicity, history of migraines, history of connective tissue disorder, active smoking, presenting AIS, fibromuscular dysplasia, single vertebral artery dissection, and multiple cervical artery vessel dissection. These were the variables that achieved statistical significance ($p<0.05$) following backward binary logistic regression including all significant variables in Table 2.

Model 3: adjusted for variables from model 1 and model 2 (age, female sex, race, Hispanic ethnicity, migraine, hypertension, diabetes, hyperlipidemia, active smoking, fibromuscular dysplasia, minor neck trauma, vertebral artery dissection, multiple vessel dissection, presenting AIS, and connective tissue disorder)

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Table 3: Univariable Analysis of Preselected Variables Across Patients With or Without *De Novo* Dissecting Aneurysm

| | All Patients (N=2938) | Patients without *de novo* DA (N=2717) | Patients with *de novo* DA(N=221) | P-value |
|---|---|---|---|---|
| Demographic | | | | |
| Age, years, median [IQR] | 45[37-55] | 45[37-55] | 45[37-54] | 0.88 |
| Female, n/N(%) | 1367/2938(46.5%) | 1260/2717(46.4%) | 107/221(48.4%) | 0.55 |
| *Race* | | | | |
| White | 2160/2938(73.5%) | 2000/2717(73.6%) | 160/221(72.4%) | 0.31 |
| Black | 148/2938(5.0%) | 140/2717(5.2%) | 8/221(3.6%) | 0.31 |
| Asian | 112/2938(3.8%) | 106/2717(3.9%) | 6/221(2.7%) | 0.31 |
| Other or unknown race | 518/2938(17.6%) | 471/2717(17.3%) | 47/221(21.3%) | 0.31 |
| Hispanic ethnicity | 231/2938(7.9%) | 224/2444(9.2%) | 7/186(3.8%) | 0.01 |
| Comorbidities | | | | |
| Migraine | 552/2938(18.8%) | 501/2717(18.4%) | 51/221(23.1%) | 0.09 |
| Hypertension | 951/2938(32.8%) | 891/2717(32.8%) | 60/221(27.1%) | 0.08 |
| Diabetes | 179/2938(6.1%) | 168/2717(6.2%) | 11/221(5.0%) | 0.47 |
| Active smoking | 537/2938(18.3%) | 509/2717(18.7%) | 28/221(12.7%) | 0.02 |
| Fibromuscular Dysplasia | 262/2938(8.9%) | 222/2713(8.2%) | 40/221(18.1%) | <0.001 |
| Connective tissue disorder | 67/2938(2.3%) | 57/2717(2.1%) | 10/221(4.5%) | 0.02 |
| Presenting Symptoms | | | | |
| Headache | 1522/2938(51.8%) | 1405/2717(51.7%) | 117/221(52.9%) | 0.72 |
| Neck Pain | 988/2938(33.6%) | 931/2717(34.3%) | 57/221(25.8%) | 0.01 |
| Minor neck trauma | 690/2938(23.5%) | 638/2717(23.5%) | 52/221(23.5%) | 0.98 |
| Presenting AIS | 1860/2938(63.3%) | 1728/2717(63.6%) | 132/221(59.7%) | 0.45 |
| Imaging | | | | |
| Dissection location | | | | |
| Single extracranial carotid dissection | 1480/2938(50.4%) | 1348/2717(49.6%) | 132/221(59.7%) | 0.005 |

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| | | | | |
|---|---|---|---|---|
| Single extracranial vertebral dissection | 987/2938(33.6%) | 936/2717(34.4%) | 51/221(23.1%) | 0.001 |
| Single intracranial carotid or vertebral dissection | 103/2938 (4.4%) | 98/2717 (3.6%) | 5/221 (2.3%) | 0.30 |
| Multiple arterial dissection | 368/2938(12.5%) | 335/2717(12.3%) | 33/221(14.9%) | 0.25 |

N=Number of Patients; IQR= interquartile range; AIS=Acute Ischemic Stroke

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Table 4: Multivariable logistic regression comparing Odds Ratios of *De Novo* Dissecting Aneurysm for Statistically Significant Variables Identified

| Variable | Unadjusted OR (95% CI, p-value) | Adjusted OR (95% CI, p-value) |
|---|---|---|
| Hispanic ethnicity | 0.38 (0.01-0.83, 0.01) | 0.40 (0.18-0.87, 0.02) |
| Active smoking | 0.62 (0.41-0.94, 0.02) | 0.62 (0.39-0.97, 0.03) |
| Connective tissue disorder | 2.21 (1.11-4.39, 0.02) | N/A (not significant following stepwise) |
| Presenting symptom – Neck Pain | 0.66 (0.48-0.91, 0.01) | 0.57 (0.40-0.81, 0.002) |
| Single carotid dissection | 1.50 (1.13-1.99, 0.005) | N/A (not significant following stepwise) |
| Single vertebral dissection | 0.57 (0.41-0.78, 0.001) | N/A (not significant following stepwise) |
| Fibromuscular dysplasia | 2.48 (1.71-3.58, <0.001) | 2.30 (1.51-3.49, <0.001) |

All variables shown in Table 4 were used for stepwise regression, only those that were statistically significant ($p<0.05$), Hispanic ethnicity, active smoking, neck pain as presenting symptom, and fibromuscular dysplasia) were used for final multivariate binary regression.

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Figure 1: Study Flow Chart

Figure 2a: Temporal trends of *de novo* dissecting aneurysm formation till last documented follow up. N=220 (1 patient with time of imaging not recorded).

Figure 2b: Temporal trends of *de novo* dissecting aneurysm formation up to 180 days after onset. N=172

Figure 3a: Kaplan-Meier Curves for ischemic stroke over 180 days in patients with baseline dissecting aneurysm vs. no baseline dissecting aneurysm. The no baseline dissecting aneurysm were also censored at the day of de novo dissecting aneurysm diagnosis if applicable.

Figure 3b: Kaplan-Meier Curves for hemorrhagic events over 180 days in patients with baseline dissecting aneurysm vs. no dissecting aneurysm. The no baseline dissecting aneurysm were also censored at the day of de novo dissecting aneurysm diagnosis if applicable.

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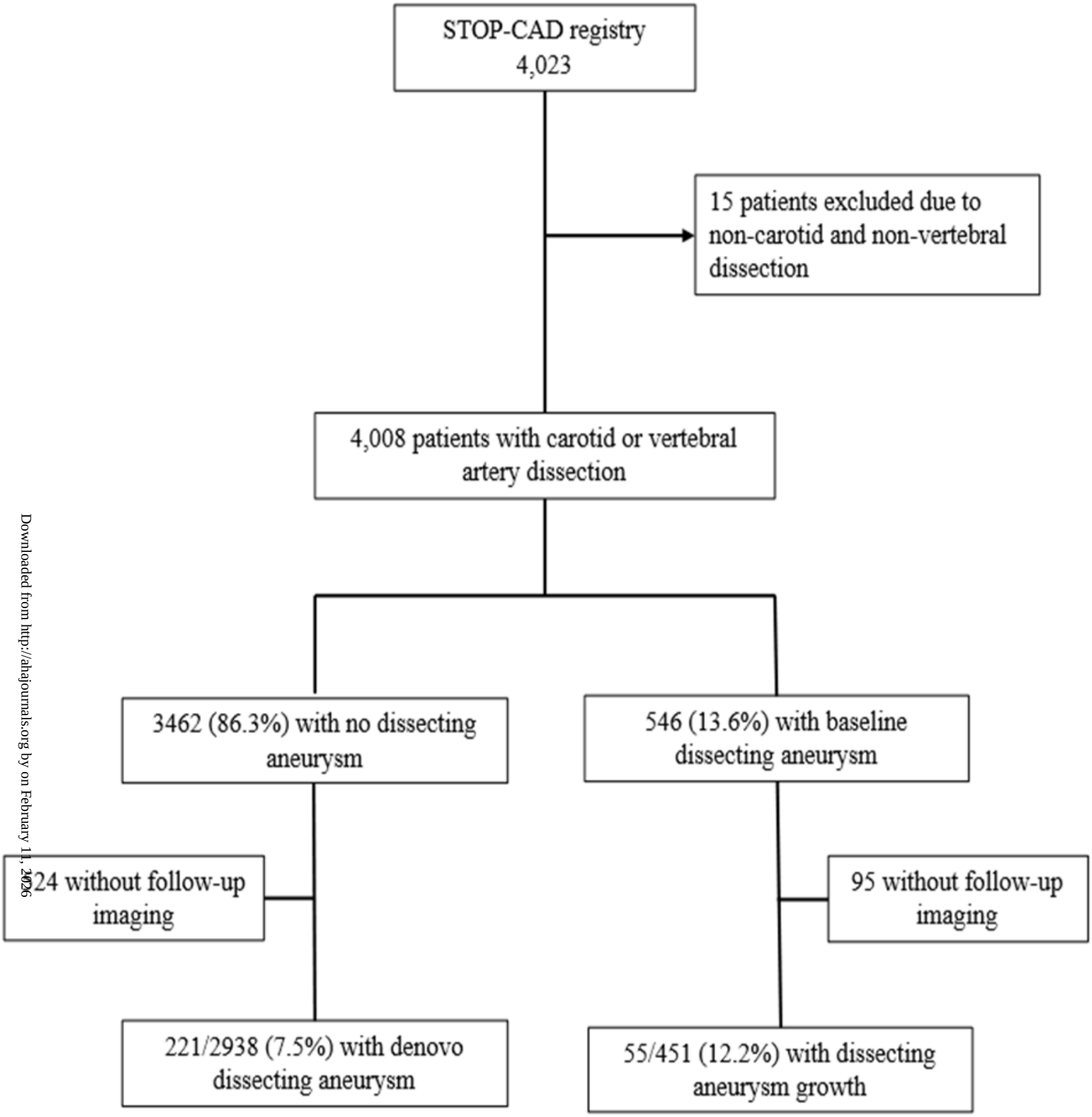
# Figure 1: Study Flowchart

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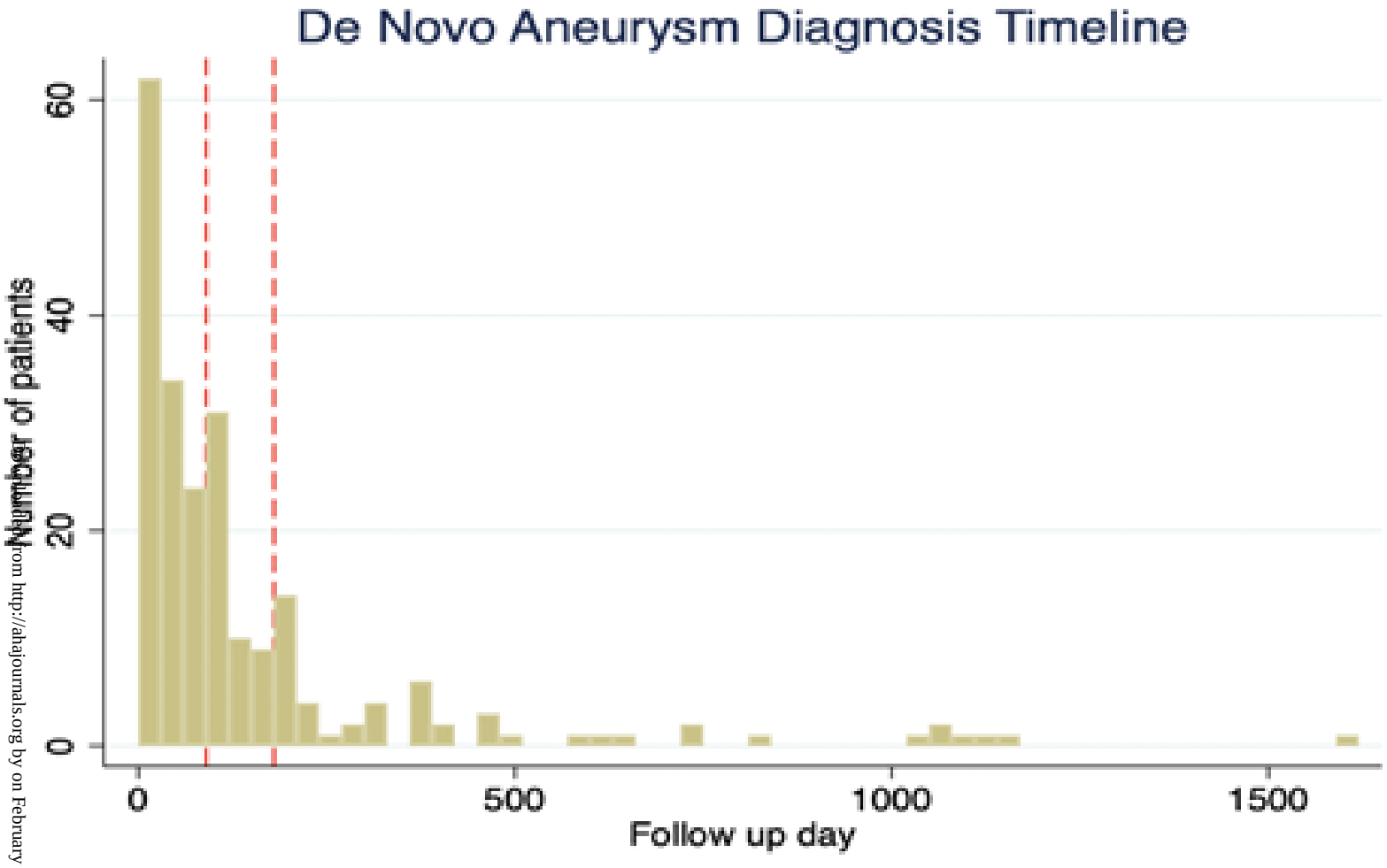


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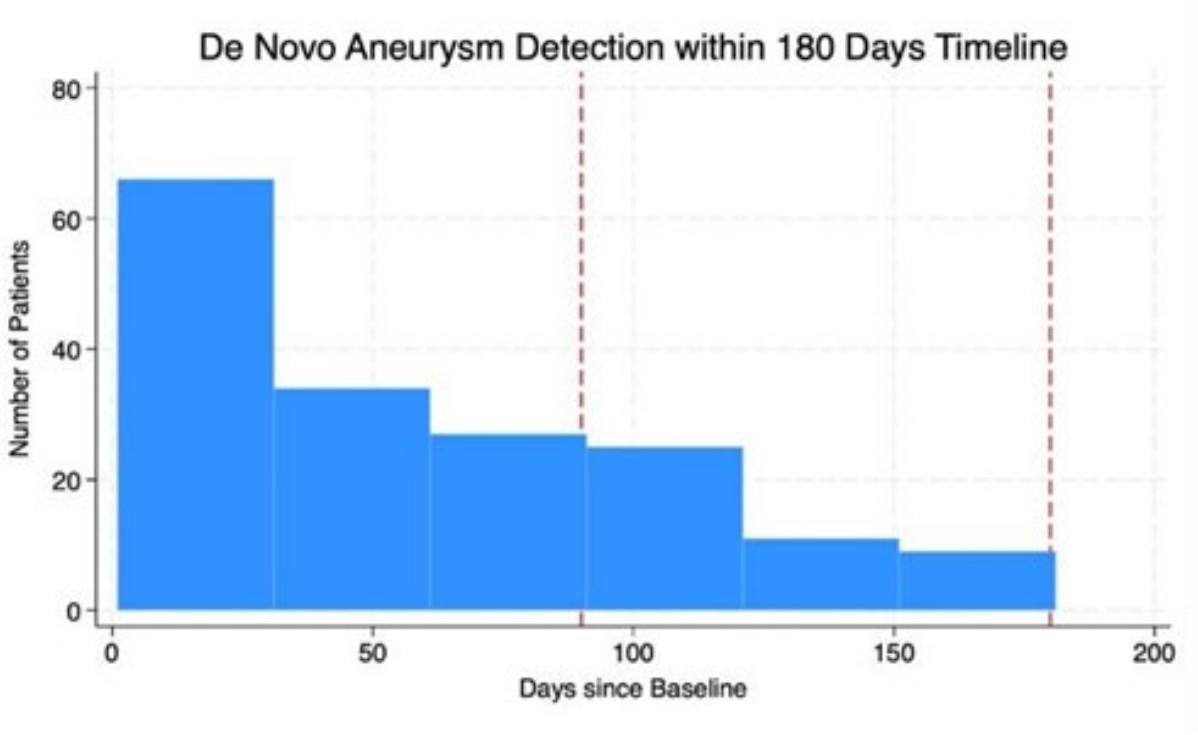


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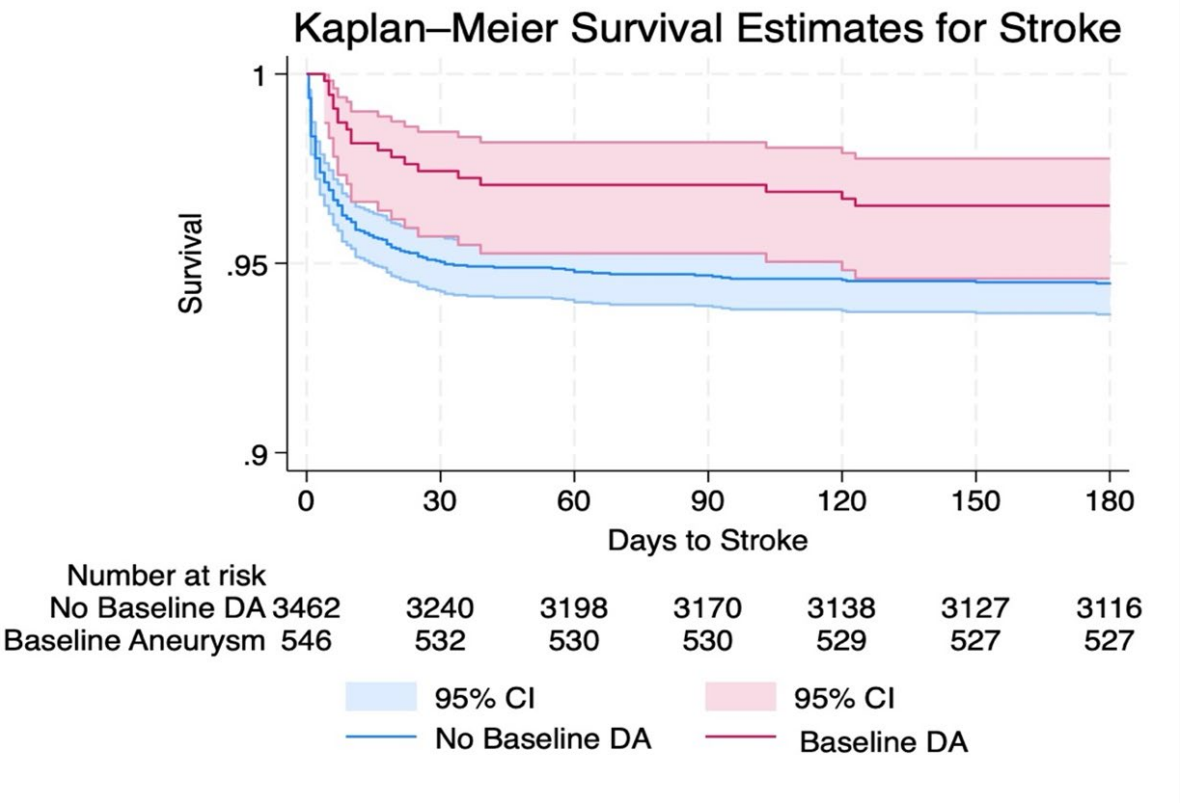


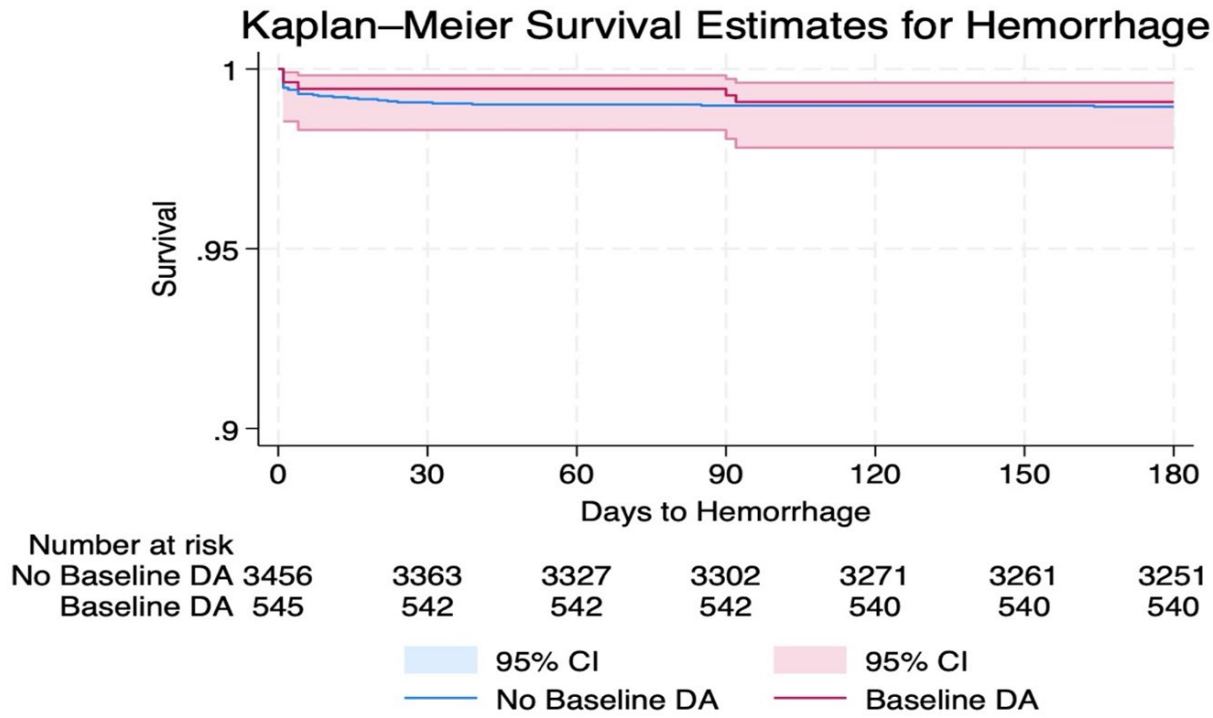


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